

Design and Analysis of a GaN-Based Megahertz Integrated Motor Drive for a PCB Motor

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Abstract—Printed circuit board (PCB) motors present a compelling alternative to commercial electric motors due to their cost-effective mass production. Furthermore, integrated motor drives have gained popularity thanks to their high power density and ease of installation. However, PCB motors face challenges due to their large effective air gap and fewer turns, resulting in low phase inductance. This article addresses these challenges by pairing the PCB motor with wide-bandgap devices and increasing the switching frequency to eliminate current ripple while reducing the size of filtering components, ultimately achieving a compact and efficient motor drive. The study explores various filtering options and decides upon for an L filter with line inductors on the motor phases. The losses of the system are analytically evaluated to a great extent leading to a selection of the switching frequency which is 1 MHz. Theoretical loss calculations are validated for both the electrical machine and inverter by means of calorimetric and electrical measurements. Finally, the integrated system is tested in motoring mode, operating at 5000 rpm and delivering 0.36 Nm of torque, resulting in an 82% efficiency. The article concludes that increasing the switching frequency is a feasible solution for low inductance motor drives.

Index Terms—Gallium nitride (GaN), integrated motor drive, loss distribution, low inductance, printed circuit board (PCB) motor, wide-bandgap.

I. INTRODUCTION

THE use of printed circuit boards (PCBs) instead of conventional wound electrical conductors in electromagnetic components has gained popularity over the years. Research into

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PCB-based magnetic components, including inductors, planar transformers, and coils for wireless power transfer, has attracted attention [1], [2], [3], [4].

In terms of rotating machinery, axial-flux permanent magnet synchronous machines (AFPMSMs) stand out to employ PCB stator technology [5], [6]. The pancake-like high-aspect-ratio composition of the AFPMSMs contributes to compactness, torque density, and power density in the design [7], [8], [9], [10]. Furthermore, the rigid composite structure of the circuit board serves a dual purpose by providing mechanical support for the stator windings [11].

Coreless printed stator designs are mainly explored in the literature due to their compatibility with existing automated manufacturing facilities, thus avoiding additional costs. As a result, in bulk orders the printed stators cost is just a fraction of other manufacturing technologies [12]. Lack of stator core reduces cogging torque and eliminates core losses on the stator. However, the trade-off is a larger effective air gap, leading to a low phase inductance [13]. This effect is elevated when comparing PCB stators to conventional copper windings, as printed boards generally have a fewer number of turns [7], [8]. An associated challenge in driving PCB AFPMSMs is the emergence of large ripple currents due to the low phase inductance [14].

In the literature, various studies capturing the excitation of motors with low winding inductance are presented, all aiming to decrease the current ripple [14], [15], [16], [17], [18], [19], [20], [21].

In [15], a small-scale ultrahigh-speed drive is developed for a motor with a phase inductance of 6.2 μH and a rated power of 100 W. The discussion concerned with the selection of appropriate drive topology comparing voltage-source inverters (VSI) and current-source inverters (CSI), both fed by a dc–dc converter to regulate the dc link voltage to reduce ripple currents, rather than choosing a higher switching frequency.

Parallely, a multilevel dc link inverter approach is discussed in [16]. The proposed inverter topology used string-connected semiconductors over the dc link to effectively adjust the output voltage again to reduce the current ripple. A three-level neutral-point-clamped inverter is utilized in [14] along with block commutation to reduce the current ripple. In a high-speed drive scenario, a similar topology is implemented with sinusoidal current output incorporating neutral point balancing techniques, as described in [17].

Another approach to consider is the use of an output filter, which can take the form of a series inductor or an inductor–capacitor (LC) filter [18], [19], [20], [21]. While this technique reduces the current ripple and attenuates electromagnetic emissions, it has certain drawbacks in terms of series voltage drop along with an increase in size, weight, and losses due to the additional filtering components.

The coextending solution can be realized with emerging wide bandgap devices such as gallium nitride (GaN) transistors. These devices have low switching and conduction losses, enabling operation at high switching frequencies. Increasing the switching frequency to the megahertz (MHz) range facilitates a feasible design utilizing a three-phase inductor in series between the motor and the drive with a considerably small size [22], [23]. In robotics and aerospace applications, not only the electrical machine nor the propriety converter power density is individually desired. In fact, the trend is to reduce the footprint as a whole to achieve a system-wise minimum volume [24]. Integrating the driver with the motor assembly in a single housing; hence the name integrated motor drive (IMD) has been applied to various systems with different power ratings [25]. For AFPMSMs, compactness is their advantage, which makes a low profile drive favorable.

In this article, an integrated drive for an AFPMSM with a PCB stator is presented. The key contribution is the integration of a low-inductance motor drive with a PCB motor optimized for efficiency and torque density in a system-level dense design. A method to limit current ripple using increased switching frequency is proposed and tested. Furthermore, the power losses in the system are investigated in detail, contributing to the selection of an optimal switching frequency.

In Section II, the PM machine is described, mechanical construction and design parameters are given. In Section III, a low-profile drive system capable of achieving switching frequencies up to 2 MHz is designed, angle sensing and control capabilities are discussed. Section IV is dedicated to a detailed analysis of losses and the integration of the motor with the drive. Section V contributes to the thermal discussions and heat management. Experimental validation and performance criteria are addressed in Section VI.

II. ELECTRICAL MACHINE DESCRIPTION

The electrical machine designed for this article is an AFPMSM with PCB stator as illustrated in Fig. 1. The machine was designed in double rotor single stator configuration which allows higher torque within a smaller volume at the expense of increased number of magnets [26], [27]. These magnets are affixed to the rotor cores while the stator has no ferromagnetic core, i.e., coreless or air cored. The absence of a stator core contributes to the elimination of core losses and reduction in the overall machine weight, thereby increases the efficiency. However, having no stator core also inherently leads to a low phase inductance. Moreover, the mechanical rigidity and manufacturing of the stator windings can be challenging problems to overcome as the stator is placed between two rotating bodies.

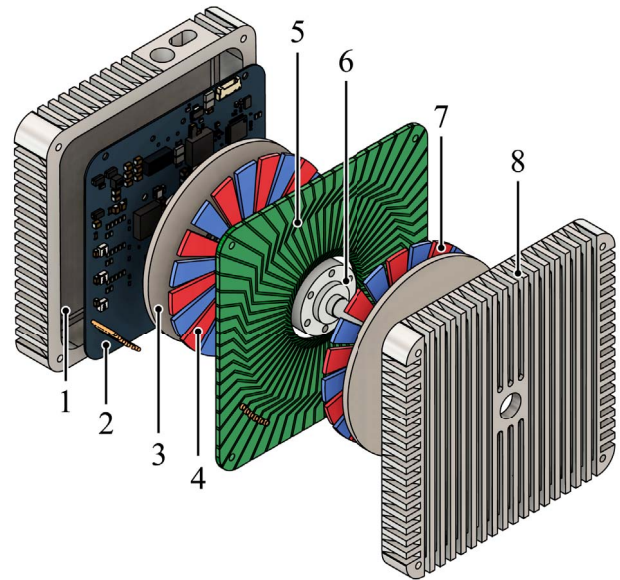


Fig. 1. Exploded view of the system. (1) Casing, (2) motor drive board, (3) rotor back-core, (4) permanent magnets, (5) PCB stator, (6) shaft, (7) reciprocal magnets, and (8) casing cover.

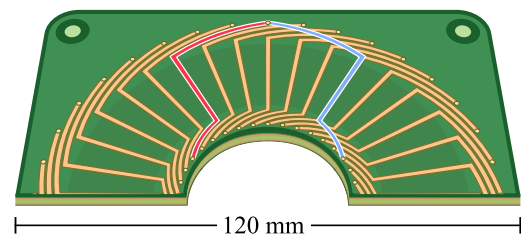


Fig. 2. Radial winding trace geometry of the PCB stator.

PCB stators offer a viable solution with ease of production and affordability through PCB manufacturing capabilities. With the windings encased within FR-4 material, it solves two problems: mechanical rigidity and electrical insulation between copper traces. Moreover, the windings of the stator can be designed in any shape, angle, and length that can be used to optimize and improve the electrical performance of the machine. A few winding designs are investigated in detail in [28]. Radial winding topology offers better electrical performance with lower phase resistances compared with other winding types. Considering the relatively low phase inductance of this winding topology in comparison to other winding designs, it becomes preferable to employ a driver design capable of accommodating this characteristic to fully exploit the motor-side advantages. The GaN-based driver and radial winding topology combine well in order to harness these features. Therefore, this article utilizes radial winding, as depicted in Fig. 2.

Overall structure of the system with driver, housings, rotors, and the stator board is presented in Fig. 1. Specifications of the PM machine are provided in Table I, with a more detailed design discussion available in [29]. The driver is connected to the stator

TABLE I
SPECIFICATIONS OF THE PCB MOTOR

Rated speed	6000 RPM	Rated frequency	1000 Hz
Rated current	4 A_{rms}	Rated torque	0.36 Nm
Phase resistance	0.36 Ω	Phase inductance	3.1 μH
Number of poles	20	Residual flux density	1.38 T
Height	120 mm	Width	120 mm
Axial length	35 mm	dc bus voltage	48 V

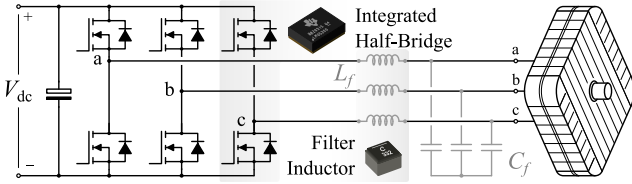


Fig. 3. Voltage source inverter topology consisting of half-bridge modules and filtering elements.

board via six pins. The system only requires terminals for the dc supply and communication signals. The housings serve several purposes. The first is that they hold whole structure together and maintain the air gap. Second, the cooling of the stator and the driver is achieved using the large surface area of the housings. Last, housings act like an electromagnetic interference (EMI) shield. The initial analysis using expected power losses and the form factor of the housing showed that the GaN drive and PCB motor can be passively cooled.

III. DESIGN OF THE INTEGRATED DRIVE SYSTEM

It is aimed to achieve precise control of both torque and speed while maintaining a high efficiency, all within a compact form factor. The proposed design operates with a voltage of 48 V, a common choice within the aerospace and automotive industries. Furthermore, it has the capability to provide the machine with an electrical power output of 230 W.

The main challenge faced by the electrical system is the low phase inductance, which results in a high current ripple issue. This study endeavors to address this problem by finding an adequate switching frequency through the utilization of high-speed GaN devices.

A. Power Electronics Description

The power stage is designed with VSI topology as depicted in Fig. 3, comprised of LMG5200 integrated half-bridge power modules from Texas Instruments, capable of functioning at frequencies of up to 10 MHz. These modules include integrated

gate drivers and can manage up to 80 V while supplying 10 A_{rms} , thanks to their low height and thermal resistance.

Several parallel multilayer ceramic capacitors (MLCCs) are used in the dc link to reduce the height of the board. Increasing the switching frequency reduces the amount of capacitance, which allows the use of ceramic capacitors that have a longer lifespan. Hence, multiple parallel MLCCs are distributed across the dc link and at each half-bridge module, resulting in a combined capacitance of 100 μF . However, it should be noted that achieving high dynamic control demands notably greater capacitance due to the stored energy in the rotor and mechanical load during transients.

The controller and power stage requires different voltage levels, therefore extra dc-dc converters are used. Due to the limited space available, the controller and the power stage are not electrically isolated, therefore a careful layout design with separate voltage rails is needed to minimize undesired coupling between.

Potential EMI and overvoltage issues are tackled by limiting the distance between the motor and drive using a short connector. Moreover, unwanted radiation is mitigated through the utilization of filters at the dc input and motor interface. A common-mode filter is employed at the input to reduce bearing currents, while the filter at the motor phases reduces the current harmonics, which will be discussed in the next section.

B. Output Filter Selection

An analysis capturing the expected current ripple is made. The aim is to investigate harmonic components in the output current that contribute to losses and efficiency reduction.

A switching function for three-phase inverter in [30] is used to analytically calculate the current ripple for sinusoidal pulse-width modulation (SPWM), which is shown in (1) at the bottom of this page. Time varying switching function, S_a , depends on the modulation m_a , carrier frequency ω_c , fundamental frequency ω_o , and time. The switching functions for all three phases are combined and multiplied by the dc link voltage in phasor domain, per phase, then divided by the motor impedance to formulate the current ripple.

Figure 4 shows that for a range between 500 kHz and 1.5 MHz, the corresponding peak current ripple over the fundamental current's peak reduces from 35% to 12%. As for the root-mean-square (rms) of the current ripple, its ratio to the rms of the fundamental current lies between 21% and 7%. Due to the inverse relationship between frequency and ripple as shown in the figure, increasing the switching frequency is not feasible in reducing the output ripple further.

Filters can be used additionally to reduce the current harmonics. Among the passive filters that can be employed, the L

$$S_a = \frac{1}{2} + \frac{m_a}{2} \cos(\omega_o t + \theta_o) + \frac{2}{\pi} \sum_{i=1}^{\infty} \left[J_o\left(i\frac{\pi}{2}m_a\right) \sin\left(i\frac{\pi}{2}\right) \cos(i\omega_c t) \right] + \frac{2}{\pi} \sum_{i=1}^{\infty} \sum_{k=-\infty}^{\infty} \left[\frac{1}{i} J_k\left(i\frac{\pi}{2}m_a\right) \sin\left(i\frac{\pi}{2} + k\frac{\pi}{2}\right) \cos((i\omega_c t) + k(\omega_o t + \theta_o)) \right]. \quad (1)$$

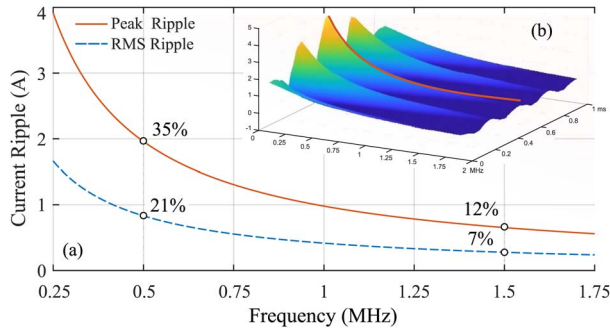


Fig. 4. (a) Expected reduction in current ripple as the switching frequency increases, without any filtering elements. (b) Current ripple change envelope in time domain for a single period.

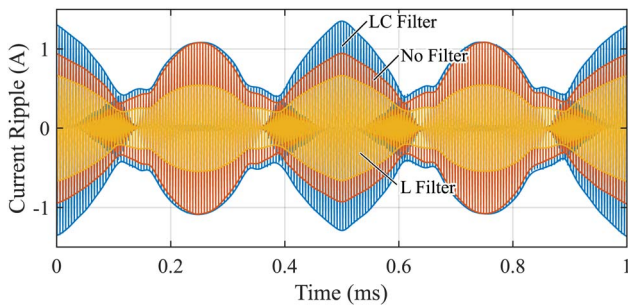


Fig. 5. Harmonics of the current ripple supplied by the inverter, excluding the fundamental frequency component, across different filter configurations, within a single fundamental period.

filter and the first-order LC filter are the most widely used. The L filter is utilized to increase the equivalent impedance seen by the high frequencies. However, the design restricts the inductor selection to low height and minimal loss criteria. Among the options that satisfy these criteria, the selected inductor comes with a height of 3.1 mm and an inductance value of $3.3 \mu\text{H}$. This inductor employs a soft core material that curbs around 65% of the inductance at 8 A. Having an inductance that is equivalent to the winding inductance, a reduction in current ripple by 50% is expected for both the motor and the inverter.

LC filter is known to yield better differential mode filtering characteristics at the output, resulting in a sinusoidal current for the motor. However, implementing a filter with a cut-off frequency set to one-tenth of the switching frequency using the specified inductor, leads to a decrease in the overall impedance seen by the inverter due to the filter capacitor (C_f). The resultant rise in the current ripple impacts the inverter, filter inductor and the filter capacitor, leading to an increase in losses over these components. To better visualize this effect, Fig. 5 compares the current ripple among the various filter options in the time domain.

At the first glance, the implementation of an LC filter appears advantageous due to its effective reduction of current harmonics in motor windings. However, this approach necessitates a filter inductor (L_f) with a higher inductance value and surface area to counterbalance the increased loss experienced in the filter and inverter. More preferred option is to employ an L filter, which permits a small current ripple to pass to the windings,

yet this results in lesser current ripple over switches and filter inductors. Consequently, this approach reduces the size of the filter, enabling a more space-efficient design. Moreover, allowing the current ripple to pass through the motor with least aggressive filter, the losses are shifted from the inverter to the windings where the heat is transferred with more ease.

Analytical discussions reveal that the air-cored PCB motor benefits more from MHz-level switching frequencies in terms of filter sizes and filter related losses. However, conventional drives with silicon-based devices would require larger inductors or LC filters in series resulting in either reduced power density or increased losses. Additionally, low inductance PCB motor coupled with a high-bandwidth GaN drive leads to improved response times, enhancing overall performance.

C. Controller and Peripherals Description

The control stage comprises the essential sensors and microcontroller required to generate the driving signals. As the level of integration increases, the number of peripherals and the control complexity also rise. This study aimed for more complete integration, which includes a central microcontroller, two current sensors, an absolute angle sensor, temperature sensors, and isolated communication interfaces.

The main controller selected is the Texas Instruments TMS320F28375D, a 32-bit dual-core microcontroller operating at 200 MHz. The current sensors, ACS711 from Allegro Microsystems, are of the hall effect type, with a frequency response that can filter high-frequency components. TMP112A temperature sensors with digital communication are employed to monitor the operating temperatures. The board utilizes communication interfaces, universal serial bus (USB) and controller area network (CAN).

An absolute angle sensor, MagAlpha MA330 from Monolithic Power Systems, is employed in an off-axis configuration with a designated encoder magnet. This setup helps reduce the axial length by up to 20%. The sensor is positioned on the same plane where the shaft protrudes, near the center hole. This side-shaft configuration eliminates the need for an additional sensor board. The sensor has a resolution of 14 bits and operates at a refresh rate of 980 kHz. Position data is collected using the serial peripheral interface (SPI), employing parallel computation to ensure the most up-to-date position information is available.

D. Printed Circuit Board Layout

The board layout is critical in enabling the various systems to work together within an integrated design. A four-layer single board is designed to minimize the height, aiming to achieve a denser overall design. Fig. 6 lays out the power, control, and communication stage boundaries with main components shown.

The driver board is positioned behind the back core of the rotor and is mechanically connected to the casing. The magnetic fields penetrating the driver are negligible because the back-iron part of the rotor acts as a magnetic shield. However, the fringing flux of the PMs can interfere with the current sensors which are placed further to the side, resulting in an angle-dependent periodic error. This issue can be resolved using a look-up-table method.

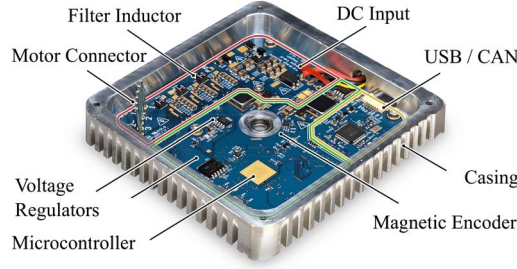


Fig. 6. Picture of the driver board mounted on motor case with various circuit blocks highlighted.

As the driver rests on the aluminum casing, large parasitic capacitances form between the whole board and the earth, which lead to common-mode currents. A common-mode filter and transient voltage suppression diodes are used to mitigate the conducted emissions and protect the circuit. Moreover, the entire system is enclosed within the casing, providing additional isolation for radiated emissions.

The board features multiple bolt holes to ensure good thermal contact between its components and the case. Additionally, thermal interface material (TIM) acts as a soft mechanic barrier for the fragile components, buffering vibrations.

The primary sources of heat over the driver board are the switches, filters, and the controller, with a total surface area of 335 mm² to the case. In contrast, the winding board has a total surface area of 6686 mm² to the case, indicating a 20-fold lower thermal resistance to the case. In Section IV, the distribution of the losses is examined in detail.

IV. LOSS ESTIMATION AND SWITCHING FREQUENCY SELECTION

Thermal design is one of the most challenging tasks in an integrated PCB motor, as compact designs often result in high loss densities. Therefore, a detailed loss analysis is crucial to ensure safe operation and identify factors that characterize the losses, such as switching frequency and current ripple.

The previous section gave an introduction to the relationship between frequency and current ripple, and subsequently, further estimations include this interconnection. The system's losses are categorized into two groups, namely electrical machine (EM) losses and inverter (INV) losses. The loss analysis calculation is formulated using analytical, experimental, and computational techniques. The contributions of these losses are compared, leading to a discussion on optimal switching frequency selection.

A. Electrical Machine Losses

These losses are generated over the electrical machine part of the motor, exclusively over the windings. Conduction loss is calculated by the well-known (2)

$$P_{EM(\text{cond}_f)} = 3I_{\text{rms}_f}^2 R_s \quad (2)$$

where I_{rms_f} is the fundamental current not considering harmonics. R_s is the measured phase resistance at 47 °C, which stay

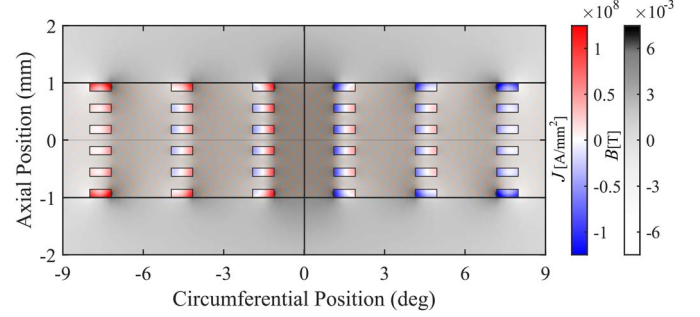


Fig. 7. One MHz FEM-simulated current density distribution and magnetic field intensity shown for PCB stator.

relatively constant for frequencies under 100 kHz. However, high frequency harmonics due to ripple current result in losses induced by proximity effect and skin effect, which are predominantly geometry-driven, making them suitable for evaluation using finite-element models (FEM). In our approach, 2-D FEM model is enough to capture the effects, due to the symmetry of the winding geometry.

A FEM model is constructed in COMSOL capturing the machine geometry, shown in Fig. 7. The model is used to evaluate various losses of the electrical machine. The conduction loss per unit length of conductor is formulated in (3)

$$P_{EM(\text{cond}_h)} = \sum_{n=1}^3 \iint_S J_n^2(f_{sw}, I_n, n) \rho_{cu} dA. \quad (3)$$

Here, n refers to the harmonic number expressed in orders of switching frequency, which is calculated for the first three components. The current density, denoted as J_n , over the conductor cross section area is influenced by several factors, including the skin effect, which is a result of the switching frequency and its harmonics and the proximity effect, related to current distribution. As the current distribution is contingent upon various factors and the interference of electric-magnetic fields, it is evaluated through finite-element analysis. The loss is estimated by integrating the current density over the conductor surface and multiplying it by the resistivity of copper (ρ_{cu}) and the total conductor length.

In air cored machines, there are no stator core losses to consider. However, machines operating at high fundamental frequencies may still encounter eddy current losses within the stator windings. Eddy current losses in stator can be described with (4)

$$P_{EM(\text{eddy}_s)} = \hat{B}_{PM}^2 f_0^2 \sigma_{cu} w_{cu}^2 V_{cu} \quad (4)$$

where $\hat{B}_{PM}$ is the amplitude of the flux density, f_0 is the fundamental frequency, σ_{cu} is the conductivity of copper, w_{cu} is the trace width, and V_{cu} is the total copper volume exposed to the magnets

$$P_{EM(\text{eddy}_r)} = \sum_{n=2}^4 \iint_S J_n^2(f_{sw}, I_n, n) \rho_{\text{steel}} dA. \quad (5)$$

Eddy current losses also occur in the rotor core ($P_{EM(\text{eddy}_r)}$) due to current harmonics. Although high-frequency harmonic

components induce an eddy current in the rotor, the magnetic field rotating at the fundamental frequency does not induce any current since the rotor rotates at the synchronous speed.

The final loss component on the electrical machine is the mechanical losses ($P_{EM(mech)}$), which includes friction and windage losses. The method of calculation for these losses is experimental, and the procedure is explained in [29].

After analyzing losses in the frequency range from 100 kHz to 2 MHz, the general trend in electrical machine losses indicate a decrease as frequency increases, primarily due to the decrease in current ripple. In addition, the resemblance between the reduction patterns of current ripple and electrical machine losses implies that current ripple has an unneglectable effect on the overall losses of the electrical machine.

B. Inverter Losses

Inverter losses include the losses produced by the power stage, control stage, and peripherals. Conduction losses for a half-bridge module is given in (6), in which the current harmonics are taken into consideration

$$P_{INV(cond)} = I_{rms}^2 R_{DS} + \sum_{n=1}^3 I_{rms(n)}^2 R_{DS}. \quad (6)$$

Here, R_{ds} is the on resistance of the GaN switches and the harmonic components are calculated up to the third order.

Switching losses are calculated for a module with well known (7)

$$P_{INV(sw)} = V_{dc} I_{rms} \frac{(t_{rise} + t_{fall})}{2} f_{sw} \quad (7)$$

where V_{dc} stands for the dc bus voltage, t_{rise} and t_{fall} denote the rise and fall times of the switches at the rated current. Both t_{rise} and t_{fall} are approximately 2 ns, which considerably helps to reduce the switching losses.

GaN transistors do not have a body diode, however there is a similar effect observed for the third quadrant operation. This effect is modelled mathematically as, $V_{SD}(\omega) = 0.45(I_{peak} \sin(\omega))^{0.39} + 1.55$ and used in the loss calculation with (8)

$$P_{INV(third)} = \int_0^{\pi/2} \frac{V_{SD}(\omega) I_{peak} \sin(\omega) t_{dead} f_{sw}}{\pi/2} d\omega. \quad (8)$$

The switching device results in higher losses for longer dead-time durations (t_{dead}). The duration, considering the frequency range and rise-fall times, is selected as 8 ns.

Driver losses are energy lost in charging and discharging the gate of the semiconductor devices in each cycle. Since the switching frequency is expected to be high, this loss is also calculated with the equation provided in (9)

$$P_{INV(driver)} = V_{gs} I_{dd}. \quad (9)$$

Here, I_{dd} is the gate driver current which increases linearly with the switching frequency as indicated in the datasheet.

Filter inductors' contribution to the loss generation is particularly important because there is no direct heat path between the

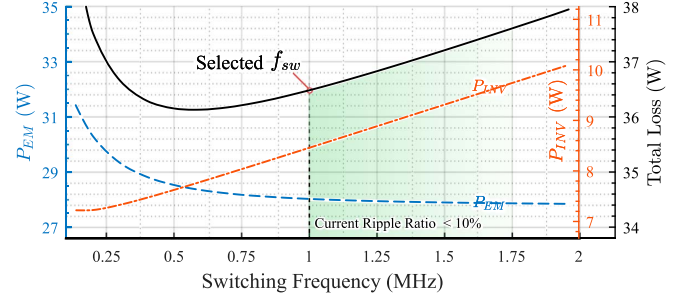


Fig. 8. Power loss trends of the inverter and electrical machine sections and the total loss curve of the system with respect to the switching frequency.

inductors and the case. The conduction loss over the inductors is given in (10)

$$P_{INV(filter)} = I_{rms}^2 R_{ind}. \quad (10)$$

Finally, the power consumption of the microcontroller and auxiliary converters, namely P_{MCU} and P_{AUX} are found experimentally.

Inverter loss analysis aimed to capture all losses, but a limitation arose in thermal validation due to the lack of detailed loss models for the line inductors. With soft magnetic cores lacking parameters and internal geometry, only direct current ohmic losses were included. While the manufacturer suggests these devices have minimal impact on motor drive operation, this limitation should be acknowledged.

Inverter losses exhibit a linear increase with an increasing switching frequency for the same range. Conversely, losses of electrical machines follow an opposite trend, indicating the existence of a minimum point in-between. Despite this point, greater attention should be paid to the inverter side since it has smaller total contact with the case.

C. Switching Frequency Selection

The appropriate switching frequency is determined by several factors, including current ripple limit, minimizing losses, and operating temperature limits.

The total power loss and operation limits are depicted in Fig. 8. A ripple ratio limit of 10% is imposed, which limits the minimum switching frequency to 1 MHz. Specifically, it is required that with selected operation frequency, the maximum allowed current ripple magnitude should not pass 10% of the peak current. The thermal limit is correlated with the loss distribution, especially for the inverter side. The intersection of these limits yields the convenient range of switching frequencies.

From an efficiency standpoint, minimizing total losses is preferable. For the designed system, total losses reach a minimum point at around 600 kHz, but the maximum deviation in total loss, within the frequency range analyzed is only 8%. This difference translates to a maximum power loss difference of approximately 2 W, with a corresponding efficiency impact of below 1%.

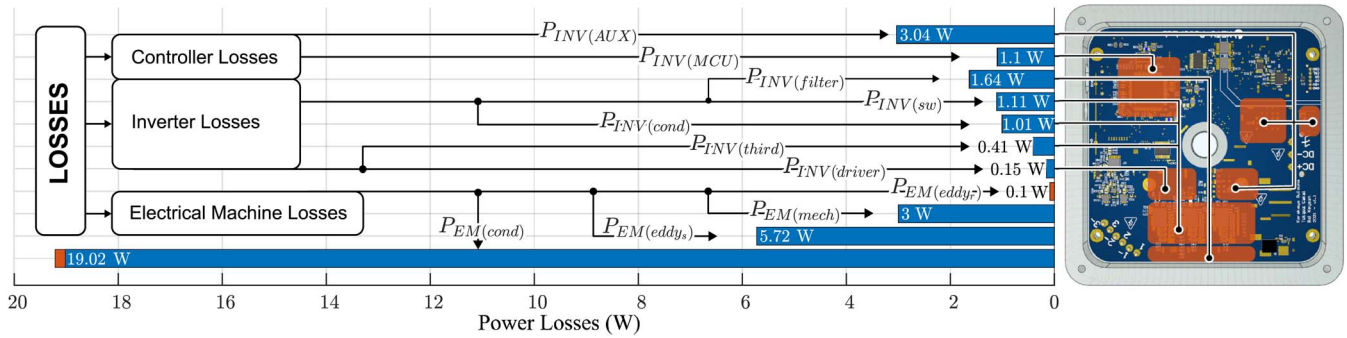


Fig. 9. Bar-tree combined graph for power losses in the system at the rated speed with 1-MHz switching frequency.

Increasing the switching frequency further reduces the current harmonics, enhance controllability, and lower computational load. However, there is a diminishing return due to the inverse relationship between frequency and ripple current. Higher switching frequencies also increase the thermal burden on the inverter because of the increased switching losses. Furthermore the small contact area with the case means that even slight increments in losses have an impact on the junction temperatures of the GaN switches.

In the light of these discussions, a switching frequency of 1 MHz is chosen. Despite not being the most efficient point of operation for the overall system, 1 MHz guarantees the current ripple requirement while avoiding the excessive thermal stresses over the driver.

Having losses laid out and variables such as switching frequency determined involves taking steps towards a loss distribution analysis, which is particularly important in an integrated system. The loss distribution of the machine is constructed for a switching frequency of 1 MHz.

Figure 9 displays a bar-tree combined graph illustrating the losses and respective components. Blue bars represent losses due to the fundamental component, while orange bars represent losses resulted from harmonics.

A major outcome of the loss analysis is the observation that a great portion of losses arises in the electrical machine section and cooled by the case. The driver losses are also directed to the case, which is now a heated medium. Despite the smaller loss budget for the driver, GaN devices excel in loss performance, validating our design choice.

The losses on the motor side are dominated by conduction and eddy current losses. The optimal trace width selection heavily depends on these two losses, as it helps to minimize the machine related losses to a great extent. Analysis of the losses indicates that the eddy current losses on the stator are quite noticeable, even with no stator core. Another notice is that, at a switching frequency of 1 MHz, the total harmonic losses only account for 0.75% of the machine losses. However, for 200 kHz the percentage is 7.5%. This suggests that for integrated drives with increased switching frequencies and non-zero current ripple, low-inductance drive design becomes more feasible.

V. THERMAL MODELING AND SIMULATION

One important aspect in the design of an integrated design is managing the heat generated. The design aims for passive cooling, therefore careful analysis is required to ensure that the cooling mechanism is sufficient.

Machine-side losses are primarily concentrated in the stator PCB, and the generated heat is assumed to be evenly distributed over the stator board. The traces run in a radial direction and are enclosed in FR-4 material. A simple volumetric average method was used to determine the equivalent thermal conductivity, which was found to be 43.4 W/(m·K). The PCB uses its outer perimeter to transfer the heat generated to the both sides of the metal case.

On the inverter side, losses are distributed across various concentration points. Fig. 9 elaborates on these points, detailing the calculated losses at each one. Specifically, controller losses are divided into two sections: one for losses over linear regulators, and the other for the microcontroller itself. In the power stage, losses are concentrated within the half-bridge modules due to conduction, switching, and third quadrant operation. These modules are top-cooled and transfer their heat to the case using thermal interface material in between packaging and the case. Although it is not a direct thermal path for the components, the driver PCB is also included in thermal modeling, with a calculated thermal conductivity for the driver board at 36 W/(m·K).

A FEM including the thermal interface materials is developed to analyze thermal behavior under rated conditions. A natural convection rate of 12 W/(m²·K) for air is used at an ambient temperature of 25 °C. The simulation results in Fig. 10 show that the case temperature reaches 46 °C, with the stator and switches reaching 52 °C and 54 °C, respectively. This demonstrates that passive cooling is a feasible thermal solution and that the output power is not constrained by thermal stresses.

VI. EXPERIMENTAL VERIFICATION

A prototype is constructed to verify the design and analysis, assessing the effectiveness of design procedures, including analytical modeling, finite-element simulations, and control

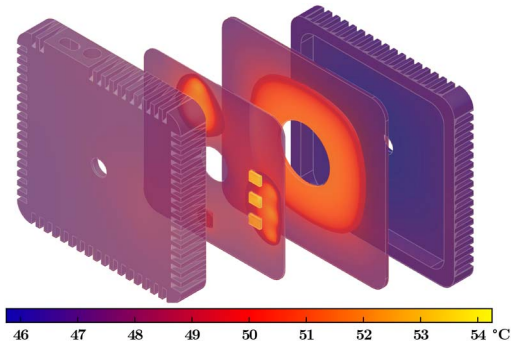


Fig. 10. Thermal FEA results with loss sources distributed according to the loss analysis.

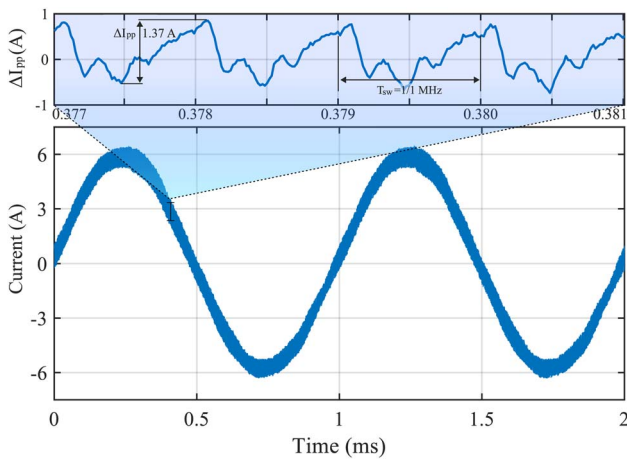


Fig. 11. Current ripple measurement using RL load with matched impedance.

methodologies. The corresponding experimental setup shown in Fig. 14 is built to validate the design and the analysis.

First, the inverter board is tested with a resistive-inductive (RL) load that matches the rated load conditions. Particularly, the inductance is matched of the assembled stator's and the voltage drop over the resistance is similar with the machine's back-EMF voltage at rated speed. The PCB stator is added in series with the load to capture the proximity and skin effects which contribute to the losses. As a consequence the operating conditions such as modulation index at 0.95 and phase current at $4 A_{rms}$ are identical to the ones at the motoring mode of operation. The current waveform observed in this experiment for the rated conditions can be seen in Fig. 11.

The current ripple value observed in RL load case closely matched the analysis at 1 MHz switching frequency. The maximum peak-to-peak and rms current ripple values and corresponding percentages are given in Table II. This test provided a verification for the output filter, creating a smooth control setting with low current harmonics.

In this setup, the impact of increasing the switching frequency on power losses and component temperatures is examined. The power losses are measured electrically, and temperature readings at various points are recorded and organized in

TABLE II
RMS AND PEAK-TO-PEAK VALUES OF THE CURRENT RIPPLE

	I_{RMS}	%	I_{pp}	%
Analytical	$0.25 A_{rms}$	6.38%	$1.32 A_{pp}$	10.45%
Experimental	$0.33 A_{rms}$	7.98%	$1.50 A_{pp}$	11.73%

TABLE III
TEMPERATURE AND LOSS TREND WITH SWITCHING FREQUENCY

F_{sw}	P_{loss}	T_{GaN}	$T_{Inductor}$	T_{case}
500 kHz	21.52 W	70 °C	85 °C	38.4 °C
750 kHz	22.84 W	73 °C	87 °C	37.0 °C
1 MHz	22.60 W	80 °C	96 °C	40.1 °C
1.25 MHz	23.32 W	90 °C	103 °C	42.5 °C
1.5 MHz	25.60 W	98 °C	105 °C	44.0 °C

Table III. The analysis include driver losses across the inverter board and conduction losses over the stator PCB. However, the eddy current losses over the stator and mechanical losses such as windage and friction are not included. Nonetheless, it's important to recognize that these unaccounted losses do not dissipate over the driver and all drive losses are properly taken into account. The experiment reveals that the temperature and power losses trend matches closely with the loss analysis outlined in Fig. 8.

The machine side losses at generator mode of operation are measured electrically, which do not include the losses due to the current ripple. The result came out to be 32 W with electrical measurement when compared with the analytical 26.4 W counterpart including the harmonics.

Additionally, using a calorimeter setup, a 9 W of power loss is measured to be dissipated over the driver at rated conditions with RL load. The analytic counterpart for the driver part estimated about 8.5 W from the bar-tree graph, which shows that they are in good agreement.

In the motoring mode, three calibrations are performed. First, the rotor is aligned with the stator by energizing a single phase with direct current, aligning the rotor with the d -axis. Next, the synchronous motor operated with volts per hertz (V/f) control. A brief position measurement was taken, assuming the motor is rotating with a constant speed. Since the position increases linearly in constant speed operation, any deviation in position captured and tabulated as error, to be removed in further measurements, correcting encoder nonlinearities. The final step involved calibrating the current sensors, addressing errors from rotor magnets' fringing flux. The motor operated in V/f control again. Upon reaching the desired speed, the switches are turned off to capture only the fringing flux field using current sensors. An error table relating to the rotor position was then created and its inverse applied to the current readings, effectively eliminating errors.

Then field-oriented-control with SPWM is used to drive the motor at 5000 rpm with 1 MHz switching frequency. Within the controller, current measurement is updated at every

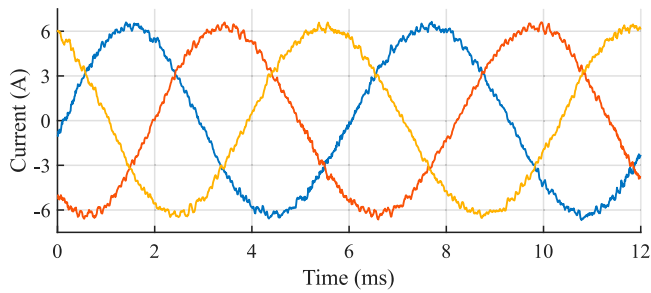


Fig. 12. Phase currents of the motor captured with integrated drive system at the rated torque and 1000 rpm.

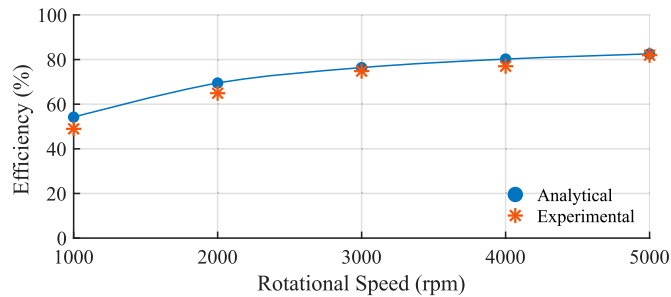


Fig. 13. Efficiency plot of the whole system with respect to the rotational speed at full load torque.



Fig. 14. Experimental setup for the motoring tests with the thermal camera view, under rated load. (1) Resistive load for the dc generator, (2) field supply for the dc generator, (3) torque display, (4) dc generator, (5) high bandwidth current probe, (6) torque sensor, (7) prototype under test, (8) oscilloscope, (9) temperature logger, and (10) dc supply.

100 kHz, and speed measurement at 10 kHz. A feed-forward compensator is implemented to increase the controller stability at high rotational speeds. Due to the system's compact integration, external measurements with an oscilloscope are not possible. The onboard current sensors are used for the visualized data in Fig. 12.

The system is tested at different mechanical loadings, the full load case resulted in 82% efficiency at 5000 rpm, 0.36 Nm. Analytic model yielding at 83.5% and hence the model and the experimental result are cooperating. An efficiency plot capturing all tested operation points is given in Fig. 13. Test setup in motoring mode with temperature readings can be seen in Fig. 14.

VII. CONCLUSION

In this article, an IMD system was designed and implemented for a low inductance axial-flux PM motor with a PCB stator, employing GaNFETs for high-frequency operation. The article suggested employing a wide bandgap device strategy to address issues related to low inductance in drive systems. A key aspect of future exploration could constitute a comparative analysis with advanced silicon-based counterparts, highlighting the strengths and weaknesses of both approaches. Furthermore, concerning controller design, future research could focus on developing an enhanced control algorithms customized to leverage the advantages offered by PCB motor and GaN-based drivers. These algorithms could optimize system performance, efficiency, and reliability by effectively utilizing the unique characteristics of these components.

The choice of filter in this article was primarily driven by considerations of efficiency and size constraints, particularly aiming to minimize current ripple experienced by the switches. The distribution of the loss components within a feasible switching frequency range was discussed, with inverter losses accounting for one-fourth of the electrical machine losses, with a remark on the inverter having a smaller surface area. A switching frequency of 1 MHz was chosen, taking into account filter selection, losses, and distribution, resulting in a reduction of harmonic components' contribution to overall losses.

Experimental results for both the electrical machine and inverter aligned well with the analytical loss model, demonstrating its suitability for evaluating high-frequency harmonic components of losses. The integrated system was successfully tested at 5000 rpm and 0.36 Nm with a 1 MHz switching frequency, affirming the viability of this approach in limiting current ripple in low inductance motor drive scenario.

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